

SOFTWARE REQUIREMENTS SPECIFICATION

Karaoke Chain Booking & POS System

Prepared as part of: AI-Assisted System Design Exercise

Version 1.0

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for the Karaoke Booking & POS System, a management system designed for a chain of karaoke lounges. The document is intended to guide the design and implementation of a backend RESTful API for the system.

1.2 Scope

The system supports three core modules: Room Booking, Order & POS (food and beverage), and Billing/Invoice. It enables customers to reserve karaoke rooms, receptionists to manage bookings and in-room orders, and branch managers to configure rooms, menu items, and view revenue reports.

Out of scope: online payment gateway integration, multi-language support, mobile application, and loyalty/membership programs. These may be considered in future iterations.

1.3 Actors

- Customer - end user who books karaoke rooms and views invoices.
- Receptionist - front-desk staff who manages bookings, check-in/out, and orders.
- Manager - branch manager who manages rooms, pricing, menu, staff, and reports.

2. Functional Requirements

2.1 Module: Room Booking

ID	Requirement	Description	Actor
FR-01	Search rooms	Search available rooms by branch, date/time range, and room type	Customer, Receptionist
FR-02	Create booking	Create a new room booking for a selected time slot	Customer, Receptionist
FR-03	Modify/cancel booking	Update or cancel an existing booking before check-in	Customer, Receptionist
FR-04	Check-in / Check-out	Mark a booking as checked-in when the customer arrives, and checked-out when the session ends	Receptionist
FR-05	Room status tracking	Track room status: Available, Booked, InUse, Cleaning	Receptionist, Manager
FR-06	View booking history	View past and upcoming bookings	Customer, Receptionist, Manager

2.2 Module: Order & POS

ID	Requirement	Description	Actor
FR-07	View menu	Browse available food and beverage menu items	Customer, Receptionist
FR-08	Create order	Add food/drink items to an active booking	Receptionist
FR-09	Update order	Modify quantity or remove items from an active order	Receptionist
FR-10	Manage menu items	Add, edit, disable menu items and prices	Manager

2.3 Module: Billing / Invoice

ID	Requirement	Description	Actor
FR-11	Generate invoice	Calculate total charge (room fee by duration + order fee) at checkout	Receptionist
FR-12	Confirm payment	Mark invoice as Paid and record payment	Receptionist
FR-13	View invoice	View invoice detail and history	Customer, Manager
FR-14	Revenue report	View aggregated revenue by branch/date range	Manager

3. User Stories

3.1 Customer

1. As a customer, I want to search for available rooms by time slot, so that I can find a suitable room to book.
2. As a customer, I want to create a booking, so that I can reserve a room in advance.
3. As a customer, I want to cancel my booking, so that I am not charged if my plan changes.
4. As a customer, I want to view my invoice after the session, so that I can verify the charges.

3.2 Receptionist

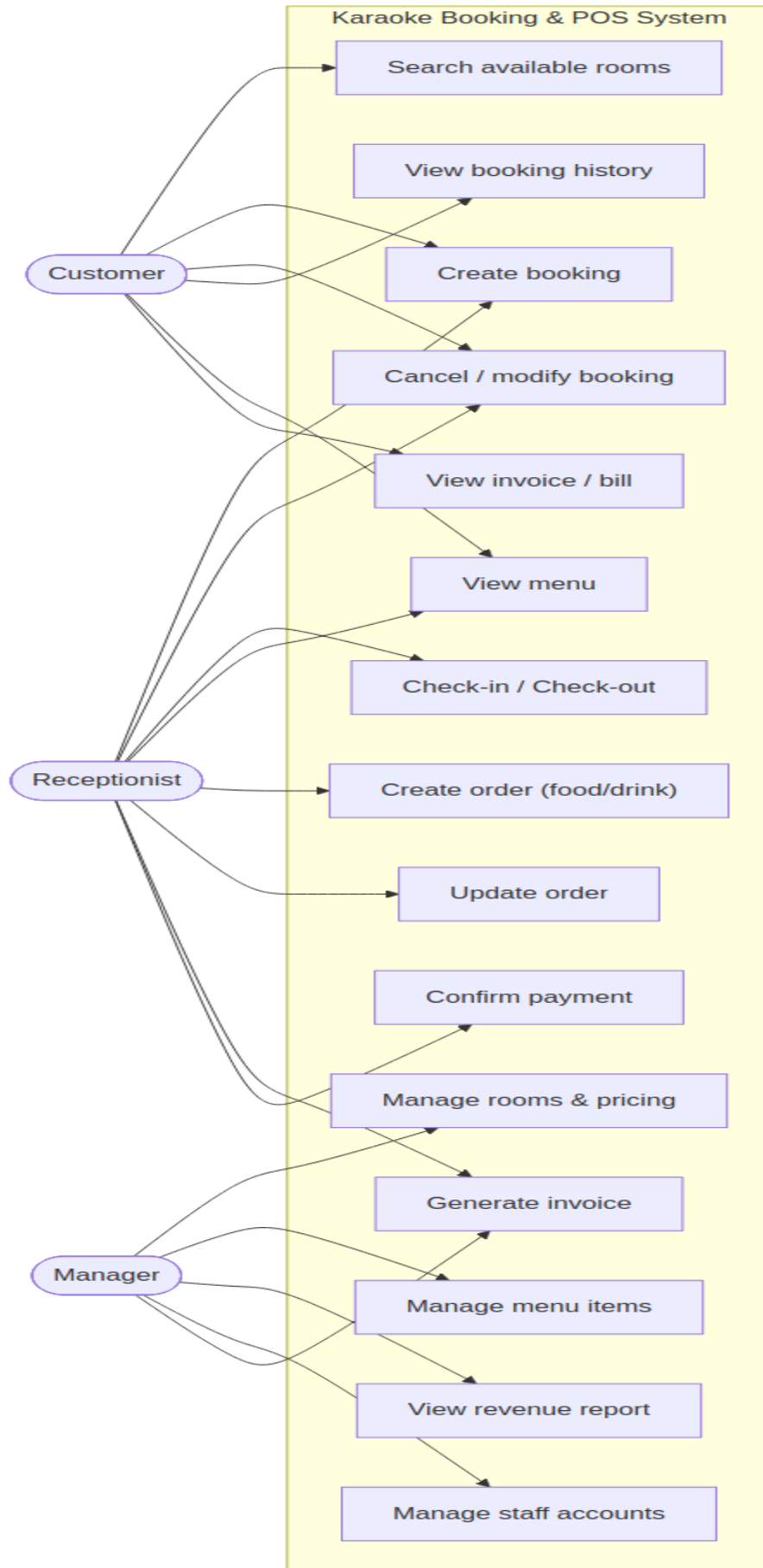
5. As a receptionist, I want to check-in a customer, so that the room status updates to InUse.
6. As a receptionist, I want to add food/drink orders to a booking, so that all charges are tracked in one place.
7. As a receptionist, I want to generate an invoice at checkout, so that the customer can pay accurately.
8. As a receptionist, I want to confirm payment, so that the booking is marked as completed.

3.3 Manager

9. As a manager, I want to manage room list and pricing, so that pricing stays up to date across branches.
10. As a manager, I want to manage menu items, so that the POS reflects current offerings.
11. As a manager, I want to view revenue reports, so that I can evaluate branch performance.
12. As a manager, I want to manage staff accounts, so that only authorized staff can access the system.

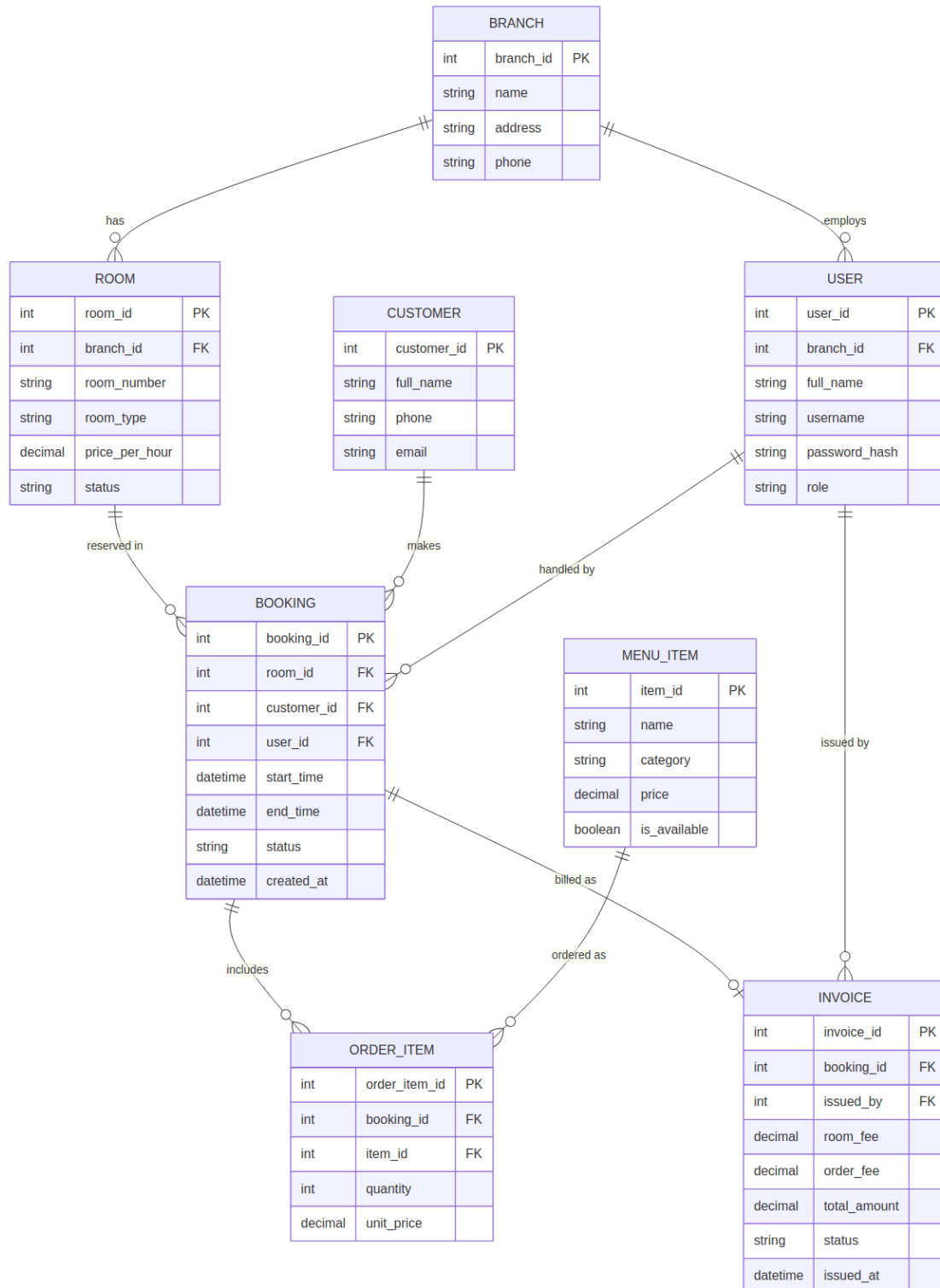
4. Use Case Diagram

The diagram below illustrates the interactions between the three actors (Customer, Receptionist, Manager) and the system use cases across the three core modules.



5. Entity Relationship Diagram (ERD)

The ERD below defines the core data entities and their relationships. Booking has a 1-to-N relationship with Room (a room can have many bookings over time) and with Order Item. The many-to-many relationship between Booking and Menu Item is resolved through the Order Item junction table.



6. Non-Functional Requirements

Category	Requirement
Performance	API responses should return within 500ms under normal load (single branch, <100 concurrent users).
Security	All endpoints except login require a valid JWT token. Passwords are stored hashed (bcrypt).
Availability	System should support 99% uptime during business hours (10:00-24:00).
Scalability	Data model supports multiple branches; architecture allows horizontal scaling of the API layer.
Usability	API responses use consistent JSON structure with clear error messages and HTTP status codes.

7. RESTful API Endpoints

7.1 Authentication

Method	Path	Description	Actor
POST	/api/auth/login	Login with username/password, returns JWT	All staff
POST	/api/auth/register	Create a staff account (Receptionist/Manager)	Manager

7.2 Room Booking Module

Method	Path	Description	Actor
GET	/api/rooms	List rooms, filter by branch/status	All
POST	/api/rooms	Create a room	Manager
PUT	/api/rooms/:id	Update room info/pricing/status	Manager, Receptionist
GET	/api/bookings	List bookings, filter by date/status	Receptionist, Manager
POST	/api/bookings	Create a booking	Customer, Receptionist
PUT	/api/bookings/:id	Update/cancel a booking	Receptionist
POST	/api/bookings/:id/checkin	Check-in a booking	Receptionist
POST	/api/bookings/:id/checkout	Check-out a booking	Receptionist

7.3 Order & POS Module

Method	Path	Description	Actor
GET	/api/menu-items	List menu items	All
POST	/api/menu-items	Create a menu item	Manager
PUT	/api/menu-items/:id	Update/disable a menu item	Manager
POST	/api/bookings/:id/orders	Add order item to a booking	Receptionist
PUT	/api/orders/:id	Update quantity of an order item	Receptionist
DELETE	/api/orders/:id	Remove an order item	Receptionist

7.4 Billing / Invoice Module

Method	Path	Description	Actor
POST	/api/bookings/:id/invoice	Generate invoice (room fee + order fee)	Receptionist
PUT	/api/invoices/:id/pay	Confirm payment, set status Paid	Receptionist
GET	/api/invoices/:id	View invoice detail	Customer, Manager, Receptionist
GET	/api/reports/revenue	Aggregated revenue by branch/date range	Manager

8. Appendix

This SRS was produced using an iterative AI-assisted prompting process. The full sequence of prompts used to generate this document and the accompanying backend source code is provided in the accompanying file History_Prompts.txt.